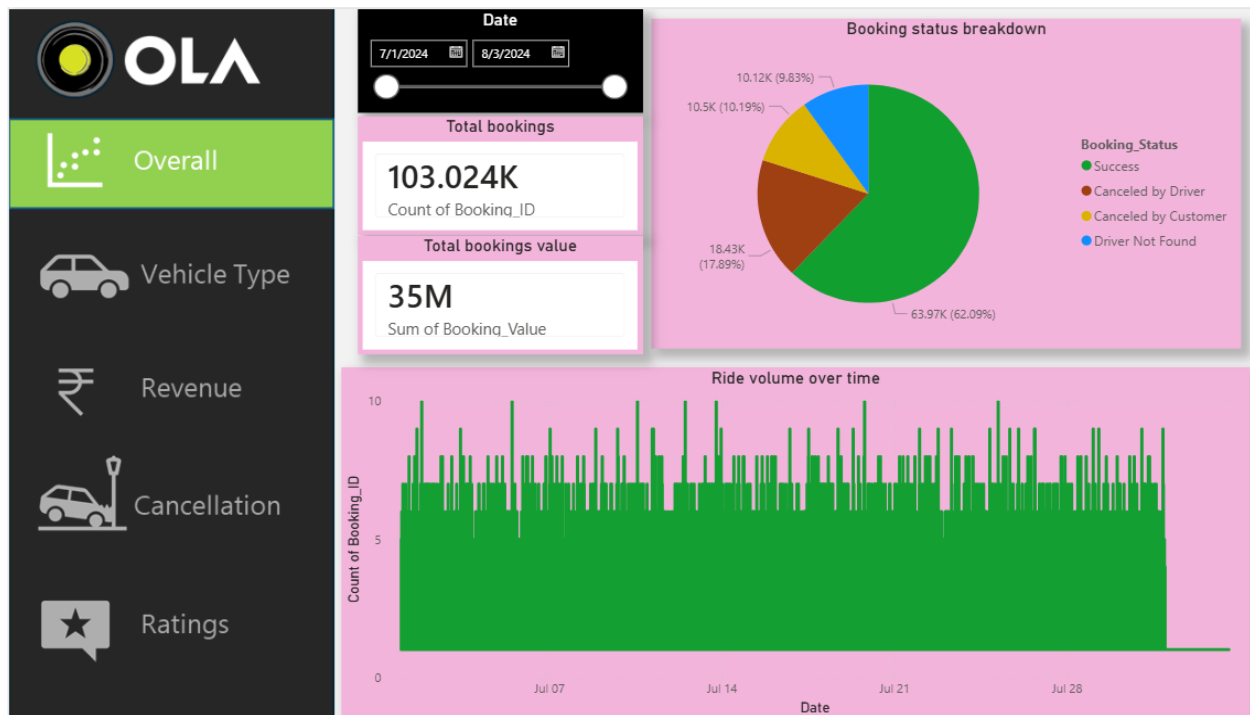


Ola Ride Analytics Dashboard | SQL + Power BI

Project Overview

Developed an interactive Ola Ride Analytics Dashboard using SQL and Power BI to analyze booking trends, revenue performance, ride cancellations, customer ratings, and vehicle-wise insights.



Problem statement :

SQL Questions:

1. Retrieve all successful bookings:
2. Find the average ride distance for each vehicle type:
3. Get the total number of cancelled rides by customers:
4. List the top 5 customers who booked the highest number of rides:
5. Get the number of rides cancelled by drivers due to personal and car-related issues:
6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
7. Retrieve all rides where payment was made using UPI:
8. Find the average customer rating per vehicle type:
9. Calculate the total booking value of rides completed successfully:
10. List all incomplete rides along with the reason:

Power BI Questions:

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

SQL work:

```
use ola ;
select count(*)
from `bookings-100000-rows`;

-- 1. Retrieve all successful bookings:

create view successful_bookings as
select * from `bookings-100000-rows`
where Booking_Status = 'success';

select * from successful_bookings;
```

Result Grid Filter Rows: Exports: Wrap Cell Contents: Fetch rows:												
Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Customer	Canceled_Rides_by_Driver	Incomplete_Ride
2024-07-27 14:00:00	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203	30	NULL	NULL	No
2024-07-28 14:00:00	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130	NULL	NULL	No
2024-07-30 14:00:00	9:02:00	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet	252	80	NULL	NULL	No
2024-07-31 14:00:00	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90	NULL	NULL	No
2024-08-01 14:00:00	9:51:00	CNR3612067560	Success	CID476071	Bike	Tumkur Road	Whitefield	133	40	NULL	NULL	No
2024-07-29 23:33:00	23:33:00	CNR4787583516	Success	CID923404	Prime Plus	Hosur Road	Jayanagar	35	55	NULL	NULL	No
2024-07-26 4:03:00	4:03:00	CNR7943634301	Success	CID647026	Prime Plus	Kammanahalli	Rajajinagar	238	95	NULL	NULL	No
2024-07-27 13:18:00	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35	NULL	NULL	No
2024-07-16 9:54:00	9:54:00	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95	NULL	NULL	No
2024-07-02 10:25:00	10:25:00	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95	NULL	NULL	No
2024-07-05 23:42:00	23:42:00	CNR3196156650	Success	CID243275	Bike	Electronic City	Langford Town	140	40	NULL	NULL	No
2024-07-09 11:11:00	11:11:00	CNR9975925287	Success	CID162055	Prime SUV	Magadi Road	RT Nagar	42	30	NULL	NULL	No
2024-07-12 14:44:00	14:44:00	CNR1591113431	Success	CID902781	eBike	Koramangala	Sarjaalur Road	245	70	NULL	NULL	No

```
-- 2. Find the average ride distance for each vehicle type:
```

```
create view ride_distance_for_each_vehicle as
select Vehicle_Type ,avg(Ride_Distance)
as avg_distance from `bookings-100000-rows`
group by Vehicle_Type;

select *from ride_distance_for_each_vehicle;
```

Result Grid Filter Rows:		
Vehicle_Type	avg_distance	
Prime Sedan	15.7649	
Bike	15.5331	
Prime SUV	15.2745	
eBike	15.5806	
Mini	15.5101	
Prime Plus	15.4475	
Auto	6.2381	

```
-- 3. Get the total number of cancelled rides by customers:
```

```
create view cancelled_rides_by_customers as
select count(*) from `bookings-100000-rows`
where Booking_Status = 'canceled by customer';
```

```
select * from cancelled_rides_by_customers;
```

Result Grid	
	count(*)
▶	10499

```
-- 4. List the top 5 customers who booked the highest number of rides:
```

```
create view top_5_customers as
select Customer_ID, count(Booking_ID) as total_rides
from `bookings-100000-rows`
group by Customer_ID
order by total_rides desc limit 5;
```

```
select * from top_5_customers ;
```

Result Grid		Filter Rows:
	Customer_ID	total_rides
▶	CID954071	5
	CID539191	4
	CID189965	4
	CID268274	4
	CID952434	4

```
-- 5. Get the number of rides cancelled by drivers due to personal and car-related issues:
```

```
create view canceled_by_driver_P_C_issues as
select count(*)
from `bookings-100000-rows`
where Canceled_Rides_by_Driver = 'Personal & Car related issue';
```

```
select * from canceled_by_driver_P_C_issues ;
```

Result Grid	
	count(*)
▶	6542

```
-- 6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
```

```
create view max_min_driver_rating_for_Prime_Sedan as
select max(Driver_Ratings) as max_rating,
min(Driver_Ratings) as min_rating
from `bookings-100000-rows` where Vehicle_Type = 'Prime Sedan';
```

```
select * from max_min_driver_rating_for_Prime_Sedan ;
```

Result Grid		Filter Rows:
	max_rating	min_rating

```
-- 7. Retrieve all rides where payment was made using UPI:
```

```
create view UPI_payment as
select * from `bookings-100000-rows`
where Payment_Method = 'UPI';

select * from UPI_payment;
```

Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Customer	Canceled_Rides_by_Driver	Incomplete_Rides
2024-07-28 14:00:00	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130	HAL	HAL	No
2024-07-31 14:00:00	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90	HAL	HAL	No
2024-07-27 13:18:00	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35	HAL	HAL	No
2024-07-16 9:54:00	9:54:00	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95	HAL	HAL	No
2024-07-02 10:25:00	10:25:00	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95	HAL	HAL	No
2024-07-09 11:11:00	11:11:00	CNR9975925287	Success	CID162055	Prime SUV	Magadi Road	RT Nagar	42	30	HAL	HAL	No
2024-07-19 21:18:00	21:18:00	CNR4443921904	Success	CID654618	Mini	Tumkur Road	Koramangala	231	50	HAL	HAL	No
2024-07-25 3:44:00	3:44:00	CNR7194303296	Success	CID538245	Mini	Mysore Road	Hennur	175	50	HAL	HAL	No
2024-07-15 17:11:00	17:11:00	CNR6494005067	Success	CID805360	Auto	Yelahanka	Malleswaram	84	60	HAL	HAL	No
2024-07-14 5:25:00	5:25:00	CNR7142279862	Success	CID378034	eBike	Yeshwanthpur	JP Nagar	210	45	HAL	HAL	No
2024-07-03 0:58:00	0:58:00	CNR5176704322	Success	CID296026	Prime Plus	KR Puram	Hennur	287	40	HAL	HAL	Yes
2024-07-10 21:56:00	21:56:00	CNR7547352327	Success	CID976231	Prime Plus	Hulimavu	Rajarajeshwa...	210	105	HAL	HAL	No
2024-07-06 15:02:00	15:02:00	CNR1568684278	Success	CID709612	Prime Plus	Bannerghatta ...	Majestic	42	90	HAL	HAL	No

```
-- 8. Find the average customer rating per vehicle type:
```

```
create view avg_customer_rating as
select Vehicle_Type, avg(Customer_Rating) as avg_customer_rating
from `bookings-100000-rows`
group by vehicle_type;

select * from avg_customer_rating;
```

Vehicle_Type	avg_customer_rating
Prime Sedan	4.001588655506982
Bike	3.993376395883525
Prime SUV	3.999377501111586
eBike	3.98785403050109
Mini	3.9977312970341075
Prime Plus	4.009498622589555
Auto	3.998810952329009

```
-- 9. Calculate the total booking value of rides completed successfully:
```

```
create view booking_value_rides_completed_successful as
select sum(Booking_Value) as total_successful_value
from `bookings-100000-rows`
where Booking_Status = 'Success';

select * from booking_value_rides_completed_successful ;
```

total_successful_value
35080467

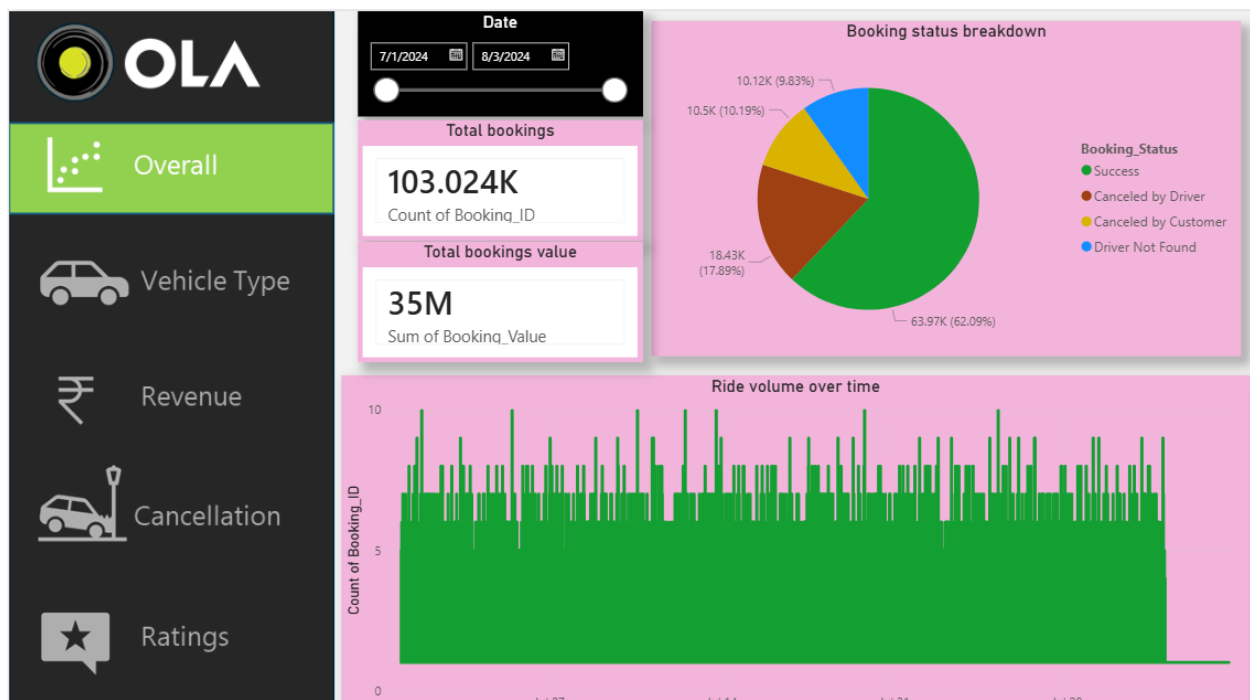
```
-- 10. List all incomplete rides along with the reason:
```

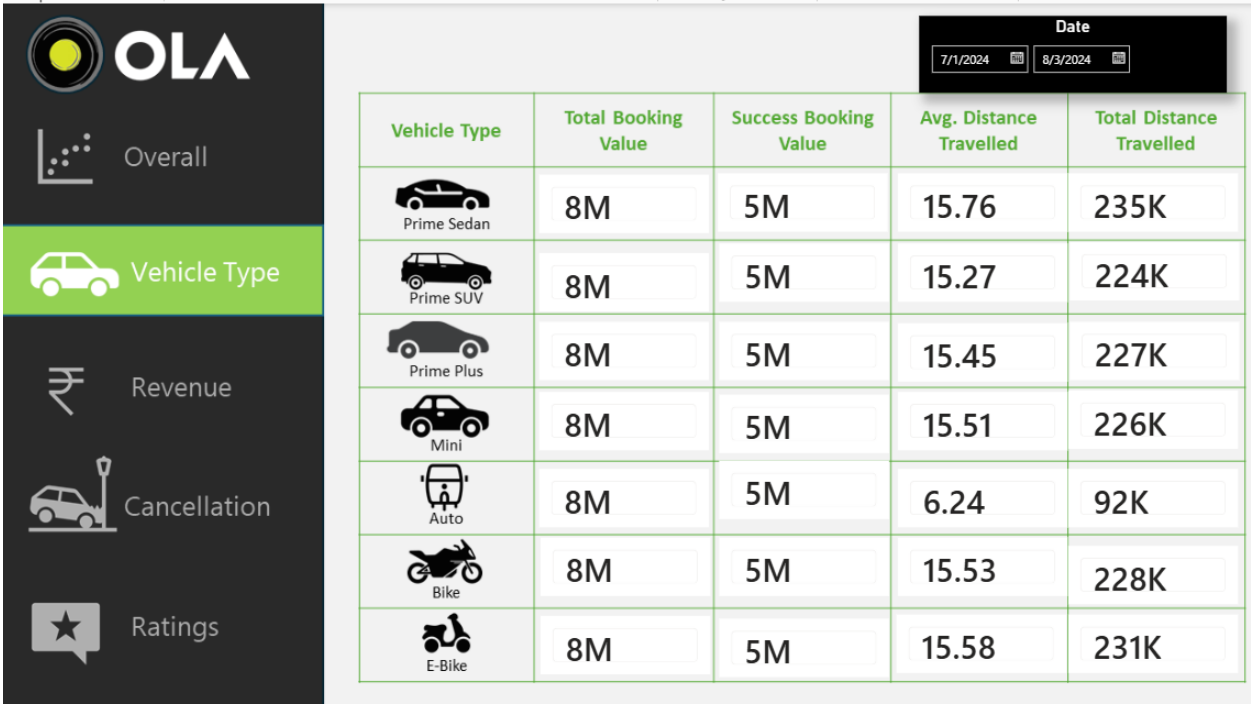
```
create view incomplete_rides as
select Booking_ID, Incomplete_Rides_Reason
from `bookings-100000-rows`
where Incomplete_Rides = 'Yes';

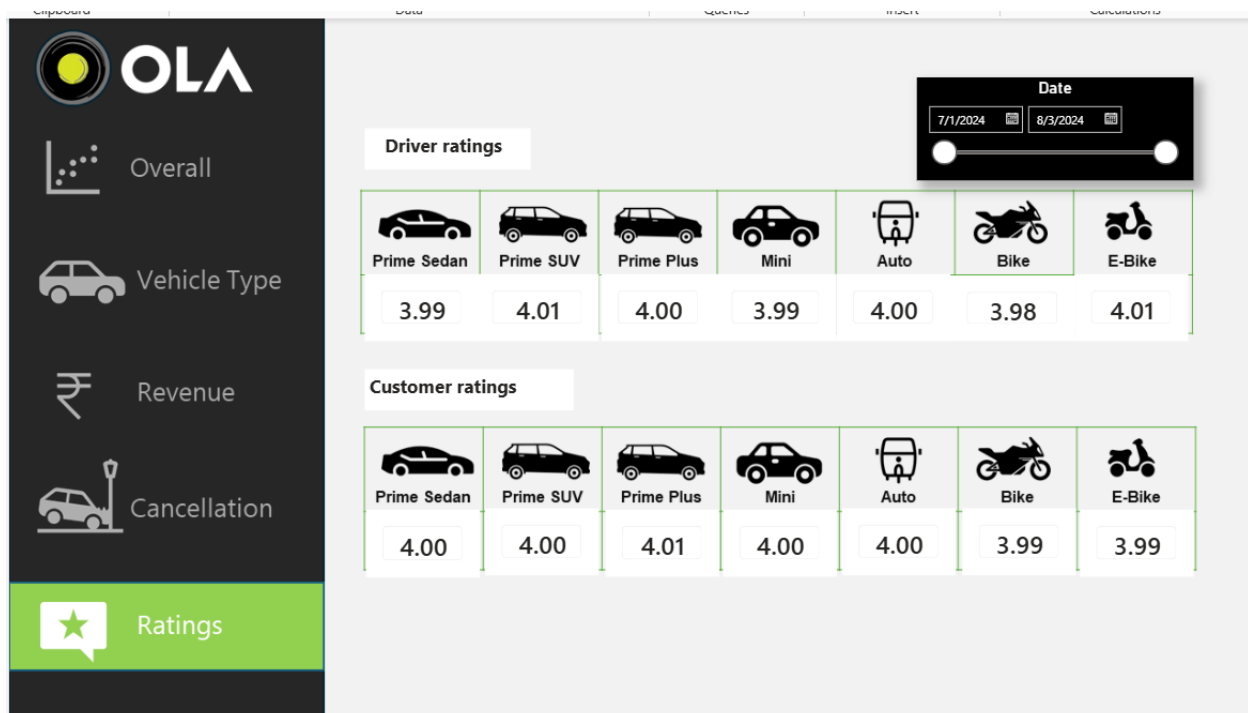
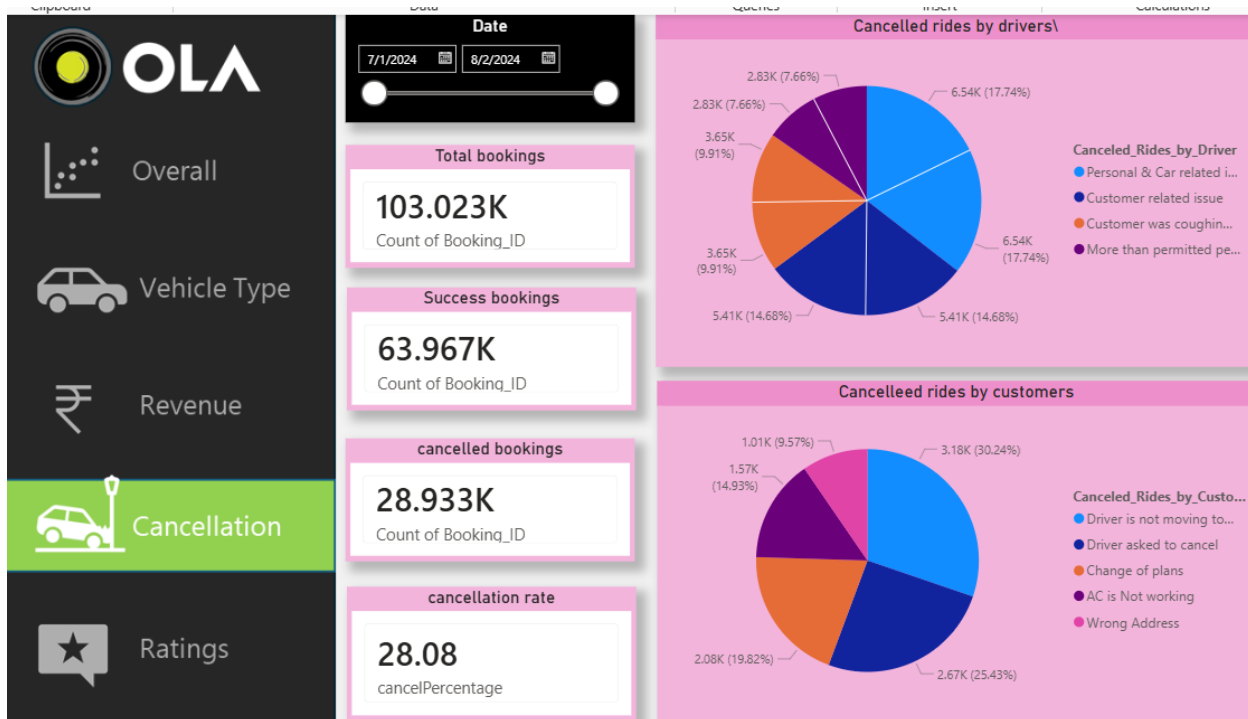
select * from incomplete_rides;
```

	Booking_ID	Incomplete_Rides_Reason
▶	CNR5176704322	Customer Demand
	CNR9312632867	Vehicle Breakdown
	CNR7924302885	Customer Demand
	CNR1640228587	Other Issue
	CNR7623690602	Other Issue
	CNR9590311980	Customer Demand
	CNR5863244684	Customer Demand
	CNR9526078867	Customer Demand
	CNR7154043084	Customer Demand
	CNR3193710797	Other Issue
	CNR7073850950	Customer Demand
	CNR9952584604	Customer Demand
	CNR5433575259	Vehicle Breakdown
	CNR3575066041	Vehicle Breakdown
	CNR7537935962	Customer Demand

PowerBI dashboard







Objectives

- * Analyze ride booking performance.**
- * Track revenue and business growth.**
- * Identify cancellation patterns.**
- * Compare vehicle category performance.**
- * Monitor customer satisfaction through ratings.**

Tools & Technologies

- * SQL (MySQL)**
- * Power BI**
- * CSV Dataset**
- * Data Cleaning & Transformation**

Key Analysis Performed

- 1. Total Bookings Analysis**
- 2. Revenue Analysis**
- 3. Vehicle Type Performance**
- 4. Cancellation Analysis**
- 5. Customer Ratings Analysis**
- 6. Interactive Dashboard Navigation**

Dashboard Pages

Overall Dashboard

- * Total Bookings**
- * Total Revenue**
- * Average Ride Value**
- * Business KPIs**

Vehicle Type Analysis

- * Vehicle-wise booking trends**
- * Revenue contribution by vehicle category**

Revenue Analysis

- * Revenue distribution**
- * High-performing ride categories**

Cancellation Analysis

- * Customer cancellations**
- * Driver cancellations**
- * Cancellation percentage trends**

Ratings Analysis

- * Customer satisfaction insights**
- * Driver performance evaluation**

Key Skills Demonstrated

- * SQL Query Writing**
- * Data Analysis**
- * Data Cleaning**
- * Dashboard Design**
- * Business Intelligence**
- * Data Visualization**
- * KPI Development**

Outcome

Built a professional business intelligence dashboard capable of transforming raw ride-booking data into actionable business insights.